

(MATH000) Mathematical Language

Notation and Conventions

Jungwoo Kim

Department of Mathematics, Pohang University of Science and Technology

Contents

Part I. General Mathematical Language	1
1 General Conventions	1
2 Theorem-like Environments	2
Part II. Notation by Mathematical Area	3
3 Set Theory and Logic	3
4 Calculus and Analysis	4
5 Linear Algebra	6
6 Differential Equations	7
7 Combinatorics and Discrete Mathematics	7
8 Probability and Statistics	8
9 Geometry	8
10 Number Theory	9
11 Algebra	9
12 Topology	10
13 Miscellaneous Notation	10

Part I. General Mathematical Language

1 General Conventions

These conventions apply throughout. When multiple notations are listed for the same concept, the first is used throughout.

Item	Convention
Scalars	Ordinary mathematical italic letters, for example a, b, c, λ, t .
Vectors	Bold lowercase letters, for example $\mathbf{u}, \mathbf{v}, \mathbf{x}$. Unless otherwise stated, vectors are column vectors.
Matrices	Bold uppercase letters, for example $\mathbf{A}, \mathbf{B}, \mathbf{M}$.
Number systems	Blackboard bold, for example $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.
Sets	Ordinary uppercase letters such as A, B, X, Y , unless a field-specific convention requires otherwise.
Families and collections	Calligraphic letters when useful, for example $\mathcal{F}, \mathcal{T}, \mathcal{B}$.
Named operators	Upright Roman type, for example $\det, \ker, \operatorname{im}, \operatorname{rk}$, and tr .
Large operators	Display size in displayed mathematics and notation tables; sums and products place indices above and below, limits place subscripts below, and integral bounds retain standard right-side placement.
Greek variants	Distinct glyphs represent distinct symbols; in particular, ε and ϵ are not interchangeable.

Notation	Meaning
$(a_1, \dots, a_n)$	Ordered tuple; parentheses are also the default matrix delimiters.
(a, b)	Open interval.
$[a, b]$	Closed interval.
$[a, b)$	Left-closed, right-open interval.
$(a, b]$	Left-open, right-closed interval.
$\{a_1, \dots, a_n\}$	Set delimiters.
$\{x \in A \mid P(x)\}$	Set-builder notation.
$\langle u, v \rangle$	Pairing or inner product, according to context.
$ x $	Absolute value, modulus, cardinality, or order, according to context.
$\ x\ $	Norm.
s.t.	Abbreviation for “such that” when naturally required.
$a = b$	Equality.
$a := b$	a is defined to be b .
$a \equiv b$	Identity, equivalence, or congruence, according to context.
$a \cong b$	Isomorphism or congruence, according to context.
$a \approx b$	Approximation.
$f: X \rightarrow Y$	Function or map from X to Y .
$x \mapsto f(x)$	Elementwise rule of a map.
$g \circ f$	Composition: first f , then g .
$f _A, f \upharpoonright_A$	Restriction of f to A .
id_X	Identity map on X .

2 Theorem-like Environments

Item	Convention
Definition	Precise meaning of a term, object, property, or construction.
Theorem	Principal mathematical result.
Proposition	Useful result of more limited scope.
Lemma	Supporting result, usually for a later proof.
Corollary	Result following readily from a preceding result.
Conjecture	Statement proposed to be true but not proved in full generality.
Hypothesis	Named assumption or proposed statement.
Formula	Distinguished computational identity or closed form.
Algorithm	Explicit step-by-step procedure.
Fact	Verified statement not requiring theorem status.
Example	Illustration of a definition, construction, phenomenon, or result.
Counterexample	Example disproving a general assertion or showing a hypothesis is necessary.
Remark	Supplementary context, warning, or qualification.
Problem	Mathematical task intended to be solved.
Open Problem	Problem not known to be solved in the stated form.
Idea	Decisive conceptual insight.
Strategy	Broad problem-solving plan.
Tactic	Local step within a larger strategy.

Proof labels are written as *Proof* and solution labels as *Solution*, each followed by a period.

Notation	Meaning
□	Ending for ordinary proofs and solutions.
■	Ending for substantial, difficult, or particularly important proofs.
Q.E.D.	Ending for exceptionally difficult, significant, or paper-level proofs.

Enumeration and Subdivision. Enumeration labels are set in upright Roman type; their form indicates the structural role of the listed items.

Item	Convention
1.1, 1.2, 1.3, ...	Independent subproblems or standalone units under a common parent.
(1), (2), (3), ...	Parallel items, such as choices, cases, statements, assumptions, or procedural steps.
(a), (b), (c), ...	Subparts of a single problem sharing a common setup.
(i), (ii), (iii), ...	Formal clauses, conditions, properties, or equivalent statements in definitions, theorems, and related statements.

Nested enumeration follows the same roles; for example, a subpart (a) may contain formal clauses (i), (ii), ...

Item	Convention
Displayed mathematics	Remains part of the surrounding sentence and receives punctuation when grammar requires it.
Equation numbering	Number equations that are referenced later or benefit from a stable label.
Logical prose	Use words such as “and” and “or” unless symbolic logic improves clarity.
Abbreviations	Use established abbreviations only when standard and unambiguous.

Part II. Notation by Mathematical Area

3 Set Theory and Logic

Uppercase letters usually denote sets; lowercase letters usually denote their elements.

Notation	Meaning
$\mathbb{N} = \{1, 2, 3, \dots\}$	Positive integers.
$\mathbb{N}_0 = \{0, 1, 2, 3, \dots\}$	Nonnegative integers.
$\mathbb{Z}$	Integers.
$\mathbb{Q}$	Rational numbers.
$\mathbb{R}$	Real numbers.
$\mathbb{C}$	Complex numbers.
$\emptyset, \varnothing$	Empty set.
$x \in A$	x is an element of A .
$x \notin A$	x is not an element of A .
$A \subseteq B$	A is a subset of B ; equality is allowed.
$A \subsetneq B$	A is a proper subset of B .
$A \supseteq B$	A is a superset of B ; equality is allowed.
$A \supsetneq B$	A is a proper superset of B .
$A \cup B$	Union.
$A \cap B$	Intersection.
$A \setminus B, A - B$	Set difference.
A^c	Complement of A in the ambient set.
$\mathcal{P}(A), 2^A$	Power set of A .
$A \times B$	Cartesian product.
$A \sqcup B$	Disjoint union.
$ A , \#A$	Cardinality.
$\{x \in A \mid P(x)\}$	Elements $x \in A$ satisfying $P(x)$.
$x \sim y$	Equivalent elements under an understood equivalence relation.
$A/\sim$	Set of equivalence classes.
$\forall x \in A$	For every $x \in A$.
$\exists x \in A$	There exists $x \in A$.
$\exists! x \in A$	There exists a unique $x \in A$.
$\nexists x \in A$	There does not exist $x \in A$.
$P \Rightarrow Q$	P implies Q .
$P \nRightarrow Q$	P does not imply Q .
$P \Leftrightarrow Q$	P if and only if Q .
$P \wedge Q$	Logical conjunction.

Notation	Meaning
$P \vee Q$	Logical disjunction.
$\neg P$	Logical negation.
$\text{dom } f$	Domain of f .
$\text{codom } f$	Codomain of f .
$f(A)$	Image of A under f .
$f^{-1}(B)$	Preimage of B under f .
$\text{im } f$	Image of the map f .
$f: X \hookrightarrow Y$	Injective map.
$f: X \twoheadrightarrow Y$	Surjective map.
$f: X \xrightarrow{\sim} Y$	Isomorphism; for sets, a bijection.
$\mathbb{1}_A(x)$	Indicator function of A .
$\aleph_0$	Cardinality of a countably infinite set.
$\mathfrak{c}$	Cardinality of the continuum.

4 Calculus and Analysis

The differential d is upright; ε is the default small positive quantity in analysis.

Notation	Meaning
$\lim_{x \rightarrow a} f(x)$	Limit as x approaches a .
$\lim_{x \rightarrow a^-} f(x)$	Left-hand limit as x approaches a .
$\lim_{x \rightarrow a^+} f(x)$	Right-hand limit as x approaches a .
$\min A$	Minimum of A .
$\max A$	Maximum of A .
$\inf A$	Infimum of A .
$\sup A$	Supremum of A .
$f'(a)$	First derivative of f at a .
$f^{(n)}(a)$	n th derivative of f at a .
$\frac{dy}{dx}$	Derivative of y with respect to x .
$\frac{\partial f}{\partial x_i}, \partial_i f$	Partial derivative with respect to x_i .
$\int_a^b f(x) \, dx$	Definite integral over $[a, b]$.
$\iint_D f \, dA$	Double integral over D .
$\iiint_\Omega f \, dV$	Triple integral over Ω .
$\oint_C f(z) \, dz$	Contour integral over a closed curve C .
$\sum_{k=1}^n a_k$	Finite sum.
$\sum_{k=1}^\infty a_k$	Infinite series.
$\prod_{k=1}^n a_k$	Finite product.
$e^x, \exp x$	Exponential function.
$\ln x, \log_e x$	Logarithm to base e .

Notation	Meaning
$\lg x, \log_{10} x$	Logarithm to base 10.
$\text{lb } x, \log_2 x$	Logarithm to base 2.
$\log_a x$	Logarithm to base a .
$\sin x$	Sine.
$\cos x$	Cosine.
$\tan x$	Tangent.
$\csc x$	Cosecant.
$\sec x$	Secant.
$\cot x$	Cotangent.
$\arcsin x, \sin^{-1} x$	Principal inverse sine.
$\arccos x, \cos^{-1} x$	Principal inverse cosine.
$\arctan x, \tan^{-1} x$	Principal inverse tangent.
$\text{arccsc } x, \csc^{-1} x$	Principal inverse cosecant.
$\text{arcsec } x, \sec^{-1} x$	Principal inverse secant.
$\text{arccot } x, \cot^{-1} x$	Principal inverse cotangent.
$\sinh x$	Hyperbolic sine.
$\cosh x$	Hyperbolic cosine.
$\tanh x$	Hyperbolic tangent.
$\text{csch } x$	Hyperbolic cosecant.
$\text{sech } x$	Hyperbolic secant.
$\coth x$	Hyperbolic cotangent.
$\text{arcsinh } x, \text{arsinh } x,$ $\sinh^{-1} x$	Principal inverse hyperbolic sine.
$\text{arccosh } x, \text{arcosh } x,$ $\cosh^{-1} x$	Principal inverse hyperbolic cosine.
$\text{artanh } x, \text{artanh } x,$ $\tanh^{-1} x$	Principal inverse hyperbolic tangent.
$\text{arcsch } x, \text{arsch } x,$ $\text{csch}^{-1} x$	Principal inverse hyperbolic cosecant.
$\text{arcsech } x, \text{arsech } x,$ $\text{sech}^{-1} x$	Principal inverse hyperbolic secant.
$\text{arcoth } x, \text{arcoth } x,$ $\coth^{-1} x$	Principal inverse hyperbolic cotangent.
$\Gamma(z)$	Gamma function.
$B(x, y)$	Beta function.
$\zeta(s)$	Riemann zeta function.
$D\mathbf{F}(\mathbf{a})$	Total derivative; Jacobian matrix in coordinates.
$\text{Jac}(\mathbf{F})$	Jacobian determinant.
$\text{Hess } f$	Hessian matrix of f .
∇f	Gradient of f .
$\nabla \cdot \mathbf{F}$	Divergence of $\mathbf{F}$.
$\nabla \times \mathbf{F}$	Curl of $\mathbf{F}$.
Δf	Laplacian of f .
$\mathbf{r}(t)$	Parametrized curve.
ds	Element of arc length.
$\int_C \mathbf{F} \cdot d\mathbf{r}$	Line integral along C .

Notation	Meaning
$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$	Flux integral through S .
$x_n \rightarrow x$	Convergence of a sequence.
$a_n \nearrow L$	Monotone increasing convergence to L .
$a_n \searrow L$	Monotone decreasing convergence to L .
$\limsup a_n$	Limit superior.
$\liminf a_n$	Limit inferior.
$f_n \rightarrow f, f_n \xrightarrow{\text{ptw}} f$	Pointwise convergence.
$f_n \Rightarrow f, f_n \xrightarrow{\text{unif}} f$	Uniform convergence.
$f_n \xrightarrow{\text{a.e.}} f$	Convergence almost everywhere.
$C^k(I)$	k times continuously differentiable functions on I .
$C^\infty(I)$	Smooth functions on I .
$\ f\ _\infty$	Supremum norm.
$B_r(x)$	Open ball of radius r centered at x .
$\overline{B}_r(x)$	Closed ball of radius r centered at x .
$z = x + iy$	Complex number with real part x and imaginary part y .
$\operatorname{Re} z$	Real part of z .
$\operatorname{Im} z$	Imaginary part of z ; distinct from lowercase im for image.
$\bar{z}$	Complex conjugate.
$ z $	Complex modulus.
$\arg z$	Argument of z .
$\operatorname{Res}(f; a), \operatorname{Res}_{z=a} f(z)$	Residue of f at a .

5 Linear Algebra

Notation	Meaning
$\mathbf{A} = (a_{ij})$	Matrix with (i, j) -entry a_{ij} .
$\mathbf{A}^\top$	Transpose of $\mathbf{A}$.
$\mathbf{A}^H$	Conjugate transpose of $\mathbf{A}$.
$\mathbf{A}^{-1}$	Inverse of $\mathbf{A}$.
$\mathbf{I}_n$	$n \times n$ identity matrix.
$\mathbf{0}$	Zero vector or zero matrix, with dimension understood.
$\mathbf{D}$	Diagonal matrix when a local name is required.
$\operatorname{diag}(d_1, \dots, d_n)$	Diagonal matrix with entries $d_1, \dots, d_n$.
$\mathbf{1}$	All-ones vector.
$\det(\mathbf{A}), \mathbf{A} $	Determinant.
$\operatorname{rk}(\mathbf{A}), \operatorname{rank}(\mathbf{A})$	Rank.
$\operatorname{tr}(\mathbf{A})$	Trace.
$\operatorname{adj}(\mathbf{A})$	Adjugate matrix.
$\operatorname{Col}(\mathbf{A})$	Column space.
$\operatorname{Row}(\mathbf{A})$	Row space.
$\operatorname{Null}(\mathbf{A}), \ker \mathbf{A}$	Null space.
$\ker T$	Kernel of a linear map T .
$\operatorname{im} T$	Image of a linear map T .
$\mathbf{e}_i$	i th standard basis vector.
$\dim V$	Dimension of V .

Notation	Meaning
$\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$	Span of the listed vectors.
$U \oplus V$	Direct sum.
$\langle \mathbf{u}, \mathbf{v} \rangle, (\mathbf{u}, \mathbf{v})$	Inner product; over complex spaces, conjugate-linear in the first argument and linear in the second.
$\mathbf{u} \cdot \mathbf{v}, \mathbf{u}^T \mathbf{v}$	Standard Euclidean inner product on $\mathbb{R}^n$.
$\mathbf{u}^H \mathbf{v}$	Standard Hermitian inner product on $\mathbb{C}^n$.
$\ \mathbf{v}\ $	Norm of $\mathbf{v}$.
$\mathbf{u} \perp \mathbf{v}$	Orthogonality.
$\mathbf{u} \times \mathbf{v}$	Cross product in $\mathbb{R}^3$.
$p_{\mathbf{A}}(\lambda), \chi_{\mathbf{A}}(\lambda)$	Characteristic polynomial of $\mathbf{A}$.
$m_{\mathbf{A}}(\lambda), \mu_{\mathbf{A}}(\lambda)$	Minimal polynomial of $\mathbf{A}$.
λ, λ_i	Eigenvalue of $\mathbf{A}$.
$\mathbf{v}, \mathbf{v}_i$	Eigenvector of $\mathbf{A}$.
$E_{\lambda}(\mathbf{A}), V_{\lambda}(\mathbf{A})$	Eigenspace of $\mathbf{A}$ associated with λ .
$\sigma(\mathbf{A}), \text{spec}(\mathbf{A})$	Spectrum of $\mathbf{A}$.
$\rho(\mathbf{A})$	Spectral radius.
$\sigma_i(\mathbf{A})$	i th singular value of $\mathbf{A}$.

6 Differential Equations

The Fourier transform uses the normalization

$$\widehat{f}(\boldsymbol{\xi}) := \int_{\mathbb{R}^n} f(\mathbf{x}) e^{-2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} d\mathbf{x}, \quad f(\mathbf{x}) = \int_{\mathbb{R}^n} \widehat{f}(\boldsymbol{\xi}) e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} d\boldsymbol{\xi},$$

whenever the Fourier inversion formula applies.

Notation	Meaning
y'	First derivative with respect to the understood independent variable.
y''	Second derivative.
$\dot{x}$	First derivative of x with respect to time.
$\ddot{x}$	Second derivative of x with respect to time.
y_h, y_c	Homogeneous or complementary part of a solution.
y_p	Particular solution.
$c_1, c_2, \dots$	Arbitrary constants in a general solution.
$W(y_1, y_2)(t), W[y_1, y_2](t)$	Wronskian of y_1 and y_2 .
$\mathcal{L}\{f(t)\}(s)$	Laplace transform of f .
$\widehat{f}(\boldsymbol{\xi}), \mathcal{F}f(\boldsymbol{\xi})$	Fourier transform under the normalization above.

7 Combinatorics and Discrete Mathematics

Notation	Meaning
$\binom{n}{r}, {}_n C_r$	Number of r -element subsets of an n -element set.
$\left(\binom{n}{r}\right), {}_n H_r$	r -combinations with repetition from n types; equals $\binom{n+r-1}{r}$.
$n!$	Factorial.

Notation	Meaning
$n^{\underline{k}}, (n)_k$	Falling factorial.
$n^{\overline{k}}, n^{(k)}$	Rising factorial.
$O(g(n))$	Big-O upper asymptotic bound.
$o(g(n))$	Little-o strict asymptotic bound.
$\Theta(g(n))$	Two-sided asymptotic order.
$\Omega(g(n))$	Asymptotic lower bound.
$G = (V, E)$	Graph with vertex set V and edge set E .
$V(G)$	Vertex set of G .
$E(G)$	Edge set of G .
$N(v), \Gamma(v)$	Neighborhood of vertex v .
$\deg(v)$	Degree of vertex v .
K_n	Complete graph on n vertices.
P_n	Path graph on n vertices.
C_n	Cycle graph on n vertices.

8 Probability and Statistics

Probability and expectation are written in blackboard bold by default.

Notation	Meaning
$\mathbb{P}(A), P(A)$	Probability of event A .
$\mathbb{P}(A \mid B), P(A \mid B)$	Conditional probability of A given B .
$X \perp\!\!\!\perp Y$	X and Y are independent.
$X \perp\!\!\!\perp Y \mid Z$	X and Y are conditionally independent given Z .
$\mathbb{E}[X], E[X]$	Expectation of random variable X .
$\mathbb{E}[X \mid Y], E[X \mid Y]$	Conditional expectation of X given Y .
$\text{Var}(X)$	Variance of X .
$\text{Cov}(X, Y)$	Covariance of X and Y .
$X \sim \text{Bin}(n, p)$	Binomial distribution.
$X \sim \text{Pois}(\lambda)$	Poisson distribution.
$X \sim \mathcal{N}(\mu, \sigma^2),$ $X \sim N(\mu, \sigma^2)$	Normal distribution with mean μ and variance σ^2 .
$X_n \xrightarrow{\text{a.s.}} X$	Almost-sure convergence.
$X_n \xrightarrow{\mathbb{P}} X$	Convergence in probability.
$X_n \xrightarrow{L^p} X$	Convergence in L^p .
$X_n \xrightarrow{d} X, X_n \Rightarrow X$	Convergence in distribution.

9 Geometry

In triangle notation, A, B, C denote vertices and a, b, c the opposite side lengths.

Notation	Meaning
$\triangle ABC$	Triangle with vertices A, B, C .
$\angle ABC$	Angle with vertex B .
$\overline{AB}$	Segment joining A and B .
AB	Length of $\overline{AB}$.
$[ABC]$	Area of $\triangle ABC$.

Notation	Meaning
$\text{length}(C), \ell(C)$	Length of C .
$\text{area}(S)$	Area of S .
$\text{vol}(S)$	Volume of S .
$\text{dist}(P, S)$	Distance from P to S .
$\text{conv}(S)$	Convex hull of S .
$\mathbb{R}^n$	n -dimensional real Euclidean space.
$\mathbb{H}^n$	n -dimensional hyperbolic space.
$\parallel$	Parallel.
$\perp$	Perpendicular or orthogonal.

10 Number Theory

Notation	Meaning
$a \mid b$	a divides b .
$a \nmid b$	a does not divide b .
$\gcd(a, b)$	Greatest common divisor, taken nonnegative.
$\text{lcm}(a, b)$	Least common multiple.
$a \equiv b \pmod{n}$	a is congruent to b modulo n .
$[a]_n$	Residue class of a modulo n .
$\mathbb{Z}/n\mathbb{Z}$	Ring of residue classes modulo n .
$(\mathbb{Z}/n\mathbb{Z})^\times$	Multiplicative group of units modulo n .
$a^{-1} \pmod{n}$	Multiplicative inverse of a modulo n .
$\left(\frac{a}{p}\right)$	Legendre symbol.
$\varphi(n)$	Euler totient function.
$\text{ord}_n(a)$	Multiplicative order of a modulo n .
$v_p(n), \nu_p(n)$	p -adic valuation of n .
$\tau(n)$	Number of positive divisors of n .
$\sigma(n)$	Sum of positive divisors of n .
$\mu(n)$	Möbius function.
$\text{rad}(n)$	Product of the distinct prime divisors of n .
$\pi(x)$	Number of primes not exceeding x .
$\mathbb{F}_q, \text{GF}(q)$	Finite field with q elements.

11 Algebra

G, H usually denote groups, R, S rings, and F, K, L fields.

Notation	Meaning
G	Group.
$e, 1_G$	Identity element of a group.
$ G $	Order of a finite group.
$\langle S \rangle$	Subgroup generated by S .
$H \leq G$	H is a subgroup of G .
$H \trianglelefteq G$	H is a normal subgroup of G .
$[G : H]$	Index of H in G .
G/H	Quotient group.

Notation	Meaning
$\ker \varphi$	Kernel of φ .
$\operatorname{im} \varphi$	Image of φ .
$\operatorname{Aut}(G)$	Automorphism group of G .
$Z(G)$	Center of G .
$C_G(x)$	Centralizer of x in G .
$\operatorname{GL}_n(F)$	General linear group over F .
$\operatorname{SL}_n(F)$	Special linear group over F .
$S_n, \mathfrak{S}_n$	Symmetric group on n symbols.
$A_n, \mathfrak{A}_n$	Alternating group on n symbols.
R	Ring.
$I \trianglelefteq R$	I is an ideal of R .
$\langle a_1, \dots, a_n \rangle, \langle a_1, \dots, a_n \rangle$	Ideal generated by $a_1, \dots, a_n$.
R/I	Quotient ring.
F	Base field in an algebraic context.
K/F	Field extension of F by K .
$F[x]$	Polynomial ring in x over F .
$\deg f$	Degree of a polynomial f .
$\operatorname{char} F$	Characteristic of F .
$\mathbb{H}$	Quaternion division algebra over $\mathbb{R}$.
$\operatorname{Gal}(L/K)$	Galois group of L/K .
$\operatorname{trdeg}_K L$	Transcendence degree of L over K .

12 Topology

Notation	Meaning
$(X, \mathcal{T})$	Topological space X with topology $\mathcal{T}$.
$A^\circ, \operatorname{int} A$	Interior of A .
$\overline{A}, \operatorname{cl} A$	Closure of A .
∂A	Boundary of A .
$\pi_1(X, x_0)$	Fundamental group of X based at x_0 .
$\chi(X)$	Euler characteristic of X .
$X \cong Y$	Homeomorphic spaces.
$X \simeq Y$	Homotopy equivalent spaces.

13 Miscellaneous Notation

Notation	Meaning
$\lfloor x \rfloor$	Floor of x .
$\lceil x \rceil$	Ceiling of x .
$\{x\}$	Fractional part $x - \lfloor x \rfloor$, when unambiguous.
$\operatorname{sgn}(x)$	Sign of a real number or permutation, according to context.
Δq	Finite change in a quantity q .
ϵ_{ijk}	Levi-Civita symbol.